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Wave energy harvester

Abstract

The disclosure relates generally to a wave energy harvester, comprising: a housing locatable aboard a floating platform; an armature coil fixedly mounted to the housing, the armature coil having a magnet associated therewith; and a body travelable along a track located within an interior of the housing, the body being coupled to the magnet; wherein, in use, wave-induced periodic motion of the floating platform results in reciprocating travel of the body along the track, with the travel of the body driving movement of the magnet with respect to the armature coil to thereby generate electricity.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a device for generating electricity. In particular, the invention

relates to a device that harvests wave energy to produce electricity.

BACKGROUND

(2) Given the growing need to reduce our carbon footprint, alternative non-fossil fuel-based energy sources are becoming increasingly important around the world. The drive to produce environmentally friendly, yet commercially viable energy has seen significant investment in clean energy technologies.

(3) The world's oceans represent a substantial potential energy source, estimated as being capable of meeting global demand twice over. Furthermore, when compared to the highly weather dependent outputs from wind and solar energy, ocean power provides a highly reliable and predictable alternative.

(4) Methods of tapping ocean energy can broadly be divided into tidal and wave technology. Whilst tidal technology relies on the potential energy related to the rise and fall of sea levels, wave energy is harvested from the kinetic energy of waves as they move along the surface of the ocean. Because wave technology provides a continuous source of energy (as opposed to the periodic output of tidal energy), wave technology is particularly attractive to coastal and island communities where it has the potential to be a mainstay renewable energy source that does not require significant storage or backup generation options.

(5) Existing wave energy harvesting devices typically require direct contact with the water, relying on a buoyancy difference between the crest and trough of a wave or the forward momentum of waves to drive an oscillating water column that extracts energy from the waves. Such devices have limited effectiveness, leading to high energy costs. Further, as many such devices are located at a significant distance from shore and/or in harsh oceanic environments, regular maintenance and damage repair is an expensive and arduous task.

(6) Accordingly, it would be advantageous to provide an improved device for harvesting wave energy. It would be beneficial to provide a device that was suitable for harvesting wave energy both close to shore and far from shore. It would be desirable if such a device was simple to maintain. The present invention was conceived with these shortcomings in mind.

SUMMARY

(7) In a first aspect, there is provided a wave energy harvester, comprising: a housing locatable aboard a floating platform; an armature coil mounted to the housing, the armature coil having a moveable magnet associated therewith; and a momentum module located within the housing, the momentum module including a body that is selectively travelable and configured to drive a corresponding movement of the magnet; wherein, in use, kinetic energy of the floating platform is transferred to the momentum module, with reciprocating travel of the body within the housing generating electricity.

(8) In a second aspect, there is provided a wave energy harvester, comprising: a housing locatable aboard a floating platform, the housing providing an interior that is isolated from a marine environment within which the floating platform is situated; an armature coil mounted to the housing, the armature coil having a moveable magnet associated therewith; and a body coupled to the magnet and selectively travelable along a track located within the interior of the housing and extending in a substantially flat plane; wherein, in use, kinetic energy from the floating platform is transferred to the body, with reciprocating travel of the body along the track driving a corresponding movement of the magnet with respect to the stationary armature coil to thereby generate electricity.

(9) An advantage provided by the present invention lies in its reliance upon the rolling motion inherent of the floating platform upon which the device is mountable to drive movement of the magnet, without the need of directly exposing the body to the marine environment.

(10) The disclosure relates generally to a wave energy harvester, comprising: a housing locatable aboard a floating platform; an armature coil fixedly mounted to the housing, the armature coil having a magnet associated therewith; and a body travelable along a track located within an interior

of the housing, the body being coupled to the magnet; wherein, in use, wave-induced periodic motion of the floating platform results in reciprocating travel of the body along the track, with the travel of the body driving movement of the magnet with respect to the armature coil to thereby generate electricity.
(11) The interior of the housing may be isolated from the marine environment within which the floating platform is situated. The housing may be watertight. Decoupling of the components within the interior of the housing from the marine environment advantageously provides a device with improved serviceability and survivability within the marine environment. Furthermore, because the armature coil and travelling body are sheltered within the housing, the present invention makes use of readily available componentry that would otherwise be unsuitable for use in such a harsh environment.
(12) In some embodiments, the magnet may be carried by the body. The movement of the magnet may be a translational movement. The traversal of the body along the track may carry the magnet through the armature coil. The linear travel of the body along the track with respect to the armature induces a current therein to thereby generate electricity through magnetic induction.
(13) In alternative embodiments, the wave energy harvester may further comprise a first engagement member carried by the body that is configured to engage with a second engagement member associated with the armature coil, the magnet being attached to the second engagement member. The movement of the magnet may be a rotational movement. The linear travel of the body along the track imparts a rotation to the second engagement member, and the resulting rotation of the magnet attached thereto induces a current within the armature coil to thereby generate electricity through magnetic induction.
(14) The track of the wave energy harvester may extend in a linear direction. The track may extend at least partly along a length of the floating platform and the body may travel along the track in response to periodic rotational motion of the floating body in pitch. Alternatively, the track may extend at least partly across a beam of the floating platform and the body may travel in response to periodic rotational motion of the floating body in roll. In a further alternative, the track may be mounted to a turntable fixed to the floating platform thereby enabling the body to travel along the track in response periodic rotational motion in both roll and pitch. Accordingly, the invention does not rely on the buoyancy provided by the difference in height between the crest and trough of a wave or on the forward momentum of the waves but instead on the continuous, and predictable roll and pitch motion that the floating platform is subjected to by the ocean waves.
(15) In some embodiments, the wave energy harvester may further include a locking element adapted to hold the body in a fixed position against gravity. The locking element may comprise a solenoid configured to release the body when the floating platform is at a peak of its periodic motion.
(16) The body may have a mass of at least 90 tonnes. A shock absorber may be located towards an end of the track. The shock absorber may be adapted to dissipate energy forces associated with stopping the body.
(17) The body may be a railway bogie. The electricity generated may be at least 180 kilowatts. By repurposing a railway bogie for use in the wave energy harvester, the need for building a bespoke body is removed, whilst providing a second use for the bogie that would otherwise be condemned to waste or scrap. This has both environmental and economic (i.e. reduced cost) benefits.
(18) The floating platform may be a floating vessel, such as a ship. Alternatively, the floating platform may be a substantially stationary floating structure, such as a concrete gravity structure. Being substantially platform agnostic, the wave energy harvester can be used in several locations, both close to and far from shore.
(19) In a further aspect, the invention provides a self-propelled vessel comprising an energy harvester as described herein. The vessel may further comprise at least one aerodynamic surface

configured to augment the wave-induced periodic motion thereof. By augmenting the wave-induced motion of the vessel, the energy generated by the wave harvester aboard can be optimised.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments of the invention are illustrated by way of example, and not by way of limitation, with reference to the accompanying drawings, of which:

(2) FIG. **1** is a side sectional view of a wave energy harvester according to an embodiment of the invention, showing a momentum module that directly moves a magnet with respect to an armature to generate electricity;

(3) FIG. **2** is a front sectional view of the harvester of FIG. **1**;

(4) FIG. **3** is a perspective view of the harvester of FIG. **1**, showing a housing enclosing the momentum module;

(5) FIG. **4** is cut-away view of FIG. **3** with the housing removed;

(6) FIG. **5** shows the harvester of FIG. **1** located aboard a deck of a ship, with rotational motion of the ship resulting in travel of the momentum module;

(7) FIG. **6** is a side sectional view of a railway bogey, the bogey forming part of the momentum module of the harvester of FIG. **1**;

(8) FIG. **7** is a top sectional view of the railway bogey of FIG. **6**;

(9) FIG. **8** is a side view of the ship of FIG. **5**, illustrating a plurality of harvesters located across the beam thereof;

(10) FIG. **9** is a top view of the ship of FIG. **8**;

(11) FIG. **10** is a plan view of a ship fitted with aerodynamic devices to augment wave induced motion;

(12) FIG. **11** is a front sectional view of the ship of FIG. **10**;

(13) FIG. **12** is a side sectional view of a wave energy harvester according to an alternative embodiment of the invention, showing a momentum module that drives movement of a magnet with respect to an armature via a rack and pinion mechanism to generate electricity;

(14) FIG. **13** is a side sectional view of an alternative embodiment of the harvester of FIG. **12**.

(15) FIG. **14** is a side sectional view of a wave energy harvester according to a further embodiment of the invention, showing a momentum module that drives movement of a magnet with respect to an armature via a hydraulic motor to generate electricity; and

(16) FIG. **15** is a side sectional view of a wave energy harvester according to a further embodiment of the invention, showing a vertically oriented momentum module that is coupled to a buoyancy device.

(17) Embodiments will now be described more fully hereinafter with reference to the accompanying drawings, in which various embodiments, although not the only possible embodiments, of the invention are shown.

DETAILED DESCRIPTION

(18) In general terms, embodiments of the invention as illustrated in the Figures relate to a wave harvesting device or harvester **10** for generating electricity. The device **10** includes a housing **11**. The housing **11** is locatable aboard a floating platform P, such as a ship or rig. The platform P is subject to periodic motion from waves within a marine environment. An armature coil **12** is fixedly mounted to the housing **11**. The armature coil **12** has a magnet **13** associated therewith. Within an interior **14** of the housing **11**, a body **15** is engaged with a track **16** to allow it to travel along the track. The periodic motion of the platform P from the waves results in reciprocating travel of the body **15** along the track **16**. The body **15** is coupled to the magnet **13**, such that travel of the body **15** along the track **16** drives a corresponding movement of the magnet **13**. Relative movement of

the magnet **13** with respect to the armature **12** results in a changing magnetic field F about the armature coil **12**. Changes in the magnet field F about the armature coil **12** induces an electrical current within the armature coil **12**, thereby generating electricity through magnetic induction.

(19) FIGS. **1** to **4** show an embodiment of the harvester **10**, in which the magnet **13** is carried by the body **15**.

(20) Best shown in FIG. **1**, the armature coil **12** is fixedly mounted to the housing **11** via supports **17**. The armature coil **12** is located substantially midway along the track **16**, between the first and second ends of the housing **11**. The armature **12** comprises a pair of tubular coils **18**, extending parallel to and in the same direction of track **16**.

(21) Together, the body **15** and track **16** form a momentum module **19**. Turning briefly to FIG. **5**, which shows the harvester **10** located aboard a deck of a tanker vessel, extending across the beam thereof. In the case of large ocean-borne platforms such as oil rigs and vessels such as oil tankers the pitch and roll of the platform experienced can be in the magnitude of 10 degrees. As the ship rolls about its length or surge axis towards the port side, the body **15** of the momentum **19** module is urged along the track **16** towards the port side. Similarly, as the ship rolls towards the starboard side, the body **15** is urged towards the starboard side. The kinetic energy of the waves is thus transferred into the momentum module **19** via the traversal of the body **15** along the track **16**.

(22) An underlying principal of the present invention is the indirect transfer of kinetic energy from the waves to the kinetic energy of the momentum module **19**. By indirect transfer, what is meant is that the momentum module **19** does not rely on direct contact with the waves and/or marine environment. Rather, the momentum module **19** relies on the inherent periodic motion of the floating platform P. As shown in FIG. **3**, the housing **11** is a watertight housing. A ventilation system comprising ventilation fans provides fluid communication between the interior **14** of the housing and the external environment, so as to maintain a controlled interior environment. The momentum module **19** is completely enclosed within the interior **14** of the housing **11**. As such the momentum module **19** is not exposed to the harsh marine environment that would otherwise be the case. Furthermore, because the momentum module **19** is located with the housing **11** aboard the platform itself, the components thereof can easily be inspected and maintained by crew and personnel, without the need for costly and potentially dangerous under-water investigations.

(23) Best shown in FIGS. **1** and **4**, the magnet **13** comprises a pair of magnetic rods, that extend through a central bore of the respective tubular coils **18** of the armature **12**. The magnetic rods are supported at either end thereof by a pair of linkages **15**. The linkages **15** are fixedly attached to the body **15**. Accordingly, as the body **15** travels along the track, the magnet **13**, carried thereby, is also driven along the track, through the stationary armature coil **12**. Movement of the magnet **13** with respect to the stationary armature **13** causes a change in the magnetic field F associated therewith, thus inducing a current within the coils **18** of the armature **12**. Accordingly, the kinetic energy of the momentum module **19** is converted into electrical energy.

(24) Turning now to FIGS. **6** and **7**, the body **15** comprises a plurality of carts **20**. The carts **20** are connected to one another such that they travel as a single unit along the track **16**. Each cart **20** is configured to carry or haul a ballast **21**. The combined mass of the loaded cart **20** can be in the order of 100 tonnes. Depending on the number of carts **20**, the total carrying capacity and hence mass of the body **15** can be altered. In the illustrated embodiment, the body **15** consists of three carts **20**, however depending on the electricity output desired, there can be a different number of carts **20**. For example, it is estimated that a body **15** having a weight of about 90 tonnes can produce a resultant electrical output of about 180 kilowatts. By increasing the mass of the body **15** (through either increasing the mass of the ballast **21** and/or the number of carts **20**) the electricity output from the harvester **10** can be increased. Each cart **20** is mechanically coupled to the track **16** via engagement members **22**. In the illustrated embodiment, the engagement members **22** are profiled wheels, rotationally mounted to opposing ends of an axle **23**.

(25) Returning to FIG. **2**. The track comprises a pair of parallel lower rails **24**, with an underside of

each wheel **22** of the body **15** being respectively engaged therewith. The lower rails **24** are fixed to a lower or bottom surface of the housing **11**, extending substantially linearly from a first end to a second end thereof. Each wheel **22** comprises a central tread portion **25** that rests against a contact surface of a respective rail **24** and an outer flange portion **26** that extends around the contact surface of the rail **24**, interlocking the wheel **22** therewith. In this embodiment, is understood that the travelling movement of the body **15** is a rolling movement. In alternative embodiments, the travelling movement can instead be a sliding movement or otherwise.
(26) The track **16** further comprises a pair of upper rails **27**. The upper rails **27** are generally parallel to the lower rails **24** such that an upper side of each wheel **22** is respectively engaged with an upper rail **27** of the track **16**. The upper rails **27** prevent the wheels **22** from lifting from the lower rails **24**, hence securing the carts **20** to the track **16**. In other embodiments (not shown), different forms of mechanical engagement between the engagement members **22** and the track **16** are contemplated. For example, the engagement members **22** can be bearings configured to run in a groove of a singular rail in a mono-rail like arrangement.
(27) In the exemplary embodiment shown, the carts **17** are railway bogies, with the respective lower **24** and upper **27** rails being regular gauge railway tracks. Accordingly, it is understood that the momentum module **19** does not require the manufacture of bespoke bodies **14** or tracks **15**. Rather, existing or decommissioned railway bogies and tracks can be repurposed for use within the harvester **10**. By repurposing or recycling existing components, the production costs associated with the harvester **10** can be reproduced, whilst, furthermore, the working life of the railway bogies is increased, reducing waste.
(28) Returning briefly to FIG. **1**. Shock absorbers **28** are disposed at either end of the track **16**. The shock absorbers are fixedly mounted to the housing **111**. The shock absorbers **28** are used to dissipate energy associated with the travelling body **15**, as the body **15** reaches the end of the track **16**.
(29) Also shown in FIG. **1**, locking elements **29** extend from a lower surface of the carts **17**. The locking elements **29** are solenoids, that magnetically couple the body **15** in a set position, against gravity. The locking elements **29** are used to release the body **15** at the maximum amplitude of the rotational motion of the platform P. In this manner, peak velocity—and hence maximal kinetic energy—of the travelling body **15** is achieved, optimising the energy output of the harvester **10**.
(30) The armature **12** is electrically coupled to a power conditioning unit **30**, which is located outside of the housing **11**. The power conditioning unit **30** is best shown in FIGS. **4** and **5**. Electricity generated by the armature **12** is transmitted to the power conditioning unit **30**. It is envisaged that the generated electricity is then transferred to a storage module such as a battery (not shown). Preferably, the battery is a liquid reduction oxidation flow battery. However it is contemplated that lithium ion or other such batteries can also be used. In other embodiments (not shown), it is contemplated that there can be more than one armature coil **12**, with each armature coil **12** being electronically connected to the or a power conditioning unit **30**.
(31) It is contemplated that multiple harvesters **10** can be fitted to a single floating platform P. FIGS. **8** and **9** shows thirteen harvesters disposed aboard the deck of a ship. As illustrated, the harvesters **10** are aligned across the beam of the ship. Accordingly, the respective momentum modules **19** of the harvesters **10** respond to rotational motions of the ship in roll. It is also understood, however, that the harvesters **10** can, alternatively, be aligned along the length or surge of the ship. In such an arrangement, the momentum modules **19** will instead respond to rotational motions of the ship in pitch, resulting from the surge of the waves. In yet another alternative, it is contemplated that the harvester is provided with a turntable (not shown) to enable the momentum module **19** to respond to rotational motions of the ship in pitch and roll. In such an arrangement, either the housing **11** can be attached to the platform P via the turntable, or the tracks **15** can be attached to the housing **11** via the turntable.
(32) By having multiple harvesters **10** fitted to a single platform P, the total output of energy can be

increased. When the harvesters **10** are located aboard a floating vessel such as a ship, it is envisaged that the energy generated can be used to power the propulsion and support systems thereon. In this way, the range of the ship is increased and the reliance on fossil fuel stores are reduced. It is envisaged that such self-propelled ships S can include wave-motion augmentation devices. With reference to FIGS. **10** and **11**, aerofoils **31** extend from a side of the ship S, and act as wings to augment roll motion of the ship S. The aerofoils **31** further include moveable control surfaces **32** in the form of ailerons. The ailerons **32** provide control to the roll motion of the ship, further contributing to the energy production of the wave energy harvester **10**. For example, when an aileron **32** fitted to the starboard side of the ship is lifted and the aileron **32** fitted to the port side is lowered, the degree of roll of the ship towards the port side will be enhanced. It is envisaged that said ailerons **32** can be controlled remotely through the use of sensors, cameras and the like, thus enabling the ship S to be a remote-controlled ship operable in rough sea conditions without the need for pilot intervention. In alternative embodiments, it is contemplated that the aerodynamic surfaces **31**, **32** can extend above the floating platform P, so as to use winds to augment motion of the platform P. In a further alternative, the surfaces can instead be hydrodynamic surfaces adapted to sit on or beneath the water and operate in a matter analogous to aerofoils **31** and ailerons **32**.

(33) Similarly, in the case of the harvesters **10** being fitted to stationary concrete gravity structures such as rigs, the electrical energy generated can be used to power the support systems aboard the rig. In both cases, the reliance on and requirement for fossil fuels is reduced or eliminated. Similarly, because the wave motion is continuous, the energy source is also continuous, as opposed to alternative renewables such as solar and wind-based systems.

(34) In addition to the harvester or harvesters **10** enabling the floating platform to become energy self-reliant, it is also contemplated that the harvester **10** and the floating platform P can together provide an electrical energy source for on-shore needs. For example, stationary rigs located close to shore can be electrically connected via transmission lines to on-shore sources, with the electricity generated by the harvesters **10** being transferred into the power grid on-shore. Alternatively, in the case of floating vessels, a concept of “energy fishing” would see the vessel travel out to sea and “fish for energy” to recharge its battery sources via the harvester **10**, before returning to dock at land where the energy from onboard battery supplies would the transferred back into the power grid. Both examples above are particularly relevant to island communities and nations whom may be resource scarce and otherwise reliant on supply of fossil fuels from neighbouring land-born countries and communities and nations. The harvester **10** would thus allow such communities to become energy self-sufficient.

(35) Moving now to FIGS. **12** and **13**, which show an alternative embodiment of the present invention, in which a momentum module **119** is used to drive a rotary generator **130** via a mechanical coupling. For clarity and ease of reference, similar reference numerals are used to describe analogous components.

(36) Wave energy harvester **110** includes momentum module **119** comprising a body **115** and track **116** as substantially described previously is housed within a housing **111**. In this embodiment, the armature coil **112** is located within a rotary generator **140**. The generator **140** that is mechanically coupled the momentum module **119**. As illustrated, the generator **140** is enclosed within an enclosure **141** that is fitted to the housing **111**. In an embodiment, the enclosure **141** is fitted on top of the housing **111**, with the generator **130** being located substantially mid-way along a length of the track **116**. This arrangement is illustrated in FIG. **12**. Alternatively, the enclosure **141** can be located at an end of the housing **111**, beyond the track **116**. This arrangement is illustrated in FIG. **13**. It is also contemplated that the generator **140** can instead be located within the housing **111** itself.

(37) The generator **140** includes magnet **113**. The magnet **113** is coupled to a shaft **142**. The shaft **142** carries an engagement member **143**. The engagement member **143** is a pinion gear. The pinion gear **113** has a toothed profile that meshes with a corresponding toothed profile of a second

engagement member **144**. The second engagement member **144** is a rack, that is fixedly connected to the body **115** of the momentum module **119**. FIG. **12** illustrates an embodiment, in which the rack **144** extends along a length of the body **115**, being attached at either end thereof to carts **120** via linkages **115**. FIG. **13** illustrates an alternative embodiment, with the rack **144** extending from one end of the body **115** to interface with the pinion **143** located at the end of the housing **111**.
(38) Accordingly, translation of the body **115** along the track **116** results in a translation of the rack **144**. In turn, the linear translation of the rack **144** imparts rotational movement upon the meshing pinion gear **113**, and, therefore, magnet **113** coupled to the shaft **142**. The relative movement of the magnet **113** with respect to the fixed armature **112** within the generator **140** results in the generation of electricity.
(39) Referring now to FIG. **14**, in a further embodiment still, a wave energy harvester **210** comprises a momentum module **219** configured to drive a hydraulic motor **250** (not shown) via at least one hydraulic cylinder **251**. Specifically, the hydraulic cylinder **251** becomes pressurised with the roll and/or pitch motion of the floating platform. In turn, the hydraulic motor imparts rotation onto a magnet coupled thereto, to thereby induce electricity within an electrical generator **240**.
(40) Moving now to another embodiment of the invention shown in FIG. **15**, in the form of wave energy harvester **310**. In this embodiment, a momentum module **319** substantially like momentum module **19** previously herein is oriented vertically. This contrasts to embodiments **10**, **110** and **210** of the wave energy harvester where the momentum module was substantially horizontal. As such, it is to be understood that momentum module **319** responds to heave motion of the floating platform, as opposed to roll and pitch.
(41) The momentum module **319** is housed within a vertical housing **311**. A buoyancy device **360**, located within the water underneath the floating platform, is connected to the momentum module **319** via a piston rod arrangement **361**. Precise alignment between the buoyancy device **360** and the momentum module **319** is not required as the connection between the piston rod arrangement **361** and the momentum module **319** is not a rigid coupling, providing off-set play therebetween. This assists in minimising construction and installation costs. A boot-like shroud **365** accords additional protection to the piston rod arrangement **361**.
(42) The buoyancy device **360** is located within a support cage **362**, which is fixedly connected to a bottom of the vertical housing **311**. The support cage **362** is provided with several bushings **363**, sliding within vertical guide rods **364** of the support cage **362** which constrain motion of the buoyancy device **360** to a substantially vertical or heave motion. Storm dampers (not shown) can be fitted to isolate the buoyancy device **360** from any rough weather encountered.
(43) Due to of the coupling of the piston rod arrangement **361**, motion of the buoyancy device **360** is followed by that of the piston rod **361** which in turn causes the translation of the weighted body **315** of the momentum module **319**, through the fixed armature coil **312** generating electrical power. When the wave has receded, and a trough is encountered, the buoyancy device **360** drops into the trough, a locking element **319** engages the weighted body **315**, retaining it in a raised position. The locking element **319** is then disengaged, allowing the weighted body **315** to fall under gravity, passing through the armature coil **312** at high speed, thereby optimising power generation. In further embodiments (not shown) it is also envisaged that the vertically mounted momentum module **319** can generate electricity via a rotary generator **340** as previously described in relation to wave energy generators **110** and **210**.
(44) It is to be understood that with the heave motion of the waves acting on the buoyancy device **360**, the wave energy harvester **310** can also be mounted to a stationary platform that is fixed to the seabed via piers. In this way, the wave energy harvester **310** relies solely on the heave motion applied to a buoyancy device **360** alone (and not in addition to the platform P) to drive the momentum module **319**.
(45) Summarily, it is understood that the harvester as described herein provides a reliable and effective device for generating electricity from wave energy. The harvester includes a momentum

module that is fully enclosed within a sealed housing and uses the roll and/or pitch motion present aboard a floating platform to drive a magnet that directly produces electricity within a stationary armature through magnetic induction. Because the momentum module is isolated from the harsh marine environment, the harvester provides a design that is easy to service and maintain without the need for costly and potentially dangerous underwater inspections. Furthermore, the use of proven and existing railway technology and components in its construction provides a robustness otherwise lacking in conventional wave energy converters, and enables the repurposing of components that would otherwise be discarded as scrap waste.
(46) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Although any methods and materials similar or equivalent to those described herein can also be used in the practice or testing of the present invention, a limited number of the example methods and materials are described herein.
(47) It is to be understood that, if any prior art publication is referred to herein, such reference does not constitute an admission that the publication forms a part of the common general knowledge in the art, in Australia or any other country.
(48) In the claims which follow and in the preceding description of the invention, except where the context requires otherwise due to express language or necessary implication, the word “comprise” or variations such as “comprises” or “comprising” is used in an inclusive sense, i.e. to specify the presence of the stated features but not to preclude the presence or addition of further features in various embodiments of the invention.
(49) TABLE-US-00001 LEGEND # No. Name 10 Wave energy harvester 11 Housing 12 Armature coil 13 Magnet 14 Interior of housing 15 Body 16 Track 17 Armature supports 18 Armature coils 19 Momentum module 20 Cart 21 Ballast 22 Engagement members 23 Axle 24 Lower rails 25 Tread portion of wheels 26 Flange portion of wheels 27 Upper rails 28 Shock absorbers 29 Locking elements 30 Power conditioning unit 31 Aerofoil 32 Aileron 110 Wave energy harvester 140 Rotary generator 141 Generator enclosure 142 Shaft 143 Pinion 144 Rack 210 Wave energy harvester 250 Hydraulic motor 251 Hydraulic cylinder 310 Wave energy harvester 360 Buoyancy device 361 Piston rod 362 Support cage 363 Bushings 364 Vertical Guide Rods 365 Shroud P Floating platform F Magnetic field S Self-propelled ship

Claims

1. A wave energy harvester, comprising: a housing locatable aboard a floating platform; an armature coil mounted to the housing, the armature coil having a moveable magnet associated therewith; and a momentum module located within the housing, the momentum module including a body that is selectively travelable and configured to drive a corresponding movement of the moveable magnet; wherein the momentum module includes a track that extends in a substantially flat plane within the housing, the body being travelable therealong; wherein, in use, kinetic energy of the floating platform is transferred to the momentum module, with reciprocating travel of the body within the housing generating electricity; wherein the track extends substantially vertically from the floating platform and the body travels in response to periodic heave motion of the floating platform.
2. The wave energy harvester of claim 1, wherein the housing provides an interior that is isolated from a marine environment within which the floating platform is situated.
3. The wave energy harvester of claim 1, wherein the moveable magnet is directly coupled to the body.
4. The wave energy harvester of claim 1, wherein the moveable magnet is carried by the body and the movement of the moveable magnet is a translational movement.
5. The wave energy harvester of claim 4, wherein the travel of the body within the housing moves

the moveable magnet through the armature coil.

6. The wave energy harvester of claim 1, wherein the moveable magnet is indirectly coupled to the body.

7. The wave energy harvester of claim 6, further comprising a first engagement member carried by the body that is configured to engage with a second engagement member associated with the armature coil, the moveable magnet being attached to the second engagement member.

8. The wave energy harvester of claim 1, wherein the track extends at least partly along a length of the floating platform and the body travels in response to periodic rotational motion of the floating body in pitch.

9. The wave energy harvester of claim 1, wherein the track extends at least partly across a beam of the floating platform and the body travels in response to periodic rotational motion of the floating body in roll.

10. The wave energy harvester of claim 1, wherein the track is mounted to a turntable fixed to the floating platform thereby enabling the body to travel in response to periodic rotational motion in both roll and pitch.

11. The wave energy harvester of claim 1, further comprising a locking element disposed within the housing and adapted to hold the body in a fixed position against gravitational forces arising from a periodic motion of the floating platform.

12. The wave energy harvester of claim 11, wherein the locking element comprises a solenoid configured to release the body when the periodic motion of the floating platform is at a peak.

13. The wave energy harvester of claim 1, further comprising a shock absorber located towards an end of the housing, the shock absorber adapted to dissipate energy forces associated with stopping the body.

14. The wave energy harvester of claim 1, wherein the body is a railway bogie.

15. A self-propelled vessel, comprising the wave energy harvester of claim 1.

16. The self-propelled vessel of claim 15, further comprising at least one control surface configured to augment wave-induced periodic motion thereof.

17. The self-propelled vessel of claim 16, wherein the at least one control surface is a moveable control surface that is operable to enhance one of roll motion and pitch motion thereof.

18. A wave energy harvester, comprising: a housing locatable aboard a floating platform; an armature coil mounted to the housing, the armature coil having a moveable magnet associated therewith; a momentum module located within the housing, the momentum module including a body that is selectively travelable and configured to drive a corresponding movement of the moveable magnet; and a shock absorber located towards an end of the housing, the shock absorber adapted to dissipate energy forces associated with stopping the body; wherein, in use, kinetic energy of the floating platform is transferred to the momentum module, with reciprocating travel of the body within the housing generating electricity.

19. The wave energy harvester of claim 18, further comprising a locking element disposed within the housing and adapted to hold the body in a fixed position against gravitational forces arising from a periodic motion of the floating platform.

20. A self-propelled vessel, comprising a wave energy harvester, wherein the wave energy harvester comprises: a housing locatable aboard a floating platform; an armature coil mounted to the housing, the armature coil having a moveable magnet associated therewith; and a momentum module located within the housing, the momentum module including a body that is selectively travelable and configured to drive a corresponding movement of the moveable magnet; wherein, in use, kinetic energy of the floating platform is transferred to the momentum module, with reciprocating travel of the body within the housing generating electricity.